Trimming Coefficients Definition

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1. Purpose

This document defines the coefficients necessary to estimate centrifugal pump characteristics modification by trimming the impeller.

2. Scope

This procedure is applicable to all centrifugal pumps.

3. Definitions

C_Q	flow coefficient for impeller trimming
C_H	head coefficient for impeller trimming
C_P	power coefficient for impeller trimming
C_η	efficiency coefficient for impeller trimming
PPC	Pump Performance Curve
N_q	pump specific speed using flowrate per impeller eye
TMP	Termomeccanica Pompe
TPI	Gabbioneta Pumps

4. General Procedure - Description

A certain size of a pump covers quite a wide range of head-flow operating conditions just depending on impeller outlet diameter. But when trimming, factoring laws cannot be used to predict pump performance modification because the original pump and the one with only outer diameter reduced are not similar. So, to transform each point of a given pump performance curve (PPC) to the corresponding one at a new diameter, it is possible to use the following formulation:

$$\frac{Q'}{Q} = \left(\frac{D'}{D}\right)^{C_Q}$$

$$\frac{H'}{H} = \left(\frac{D'}{D}\right)^{C_H}$$

$$\frac{\eta'}{\eta} = \left(\frac{D'}{D}\right)^{C_\eta}$$

$$\frac{P'}{P} = \left(\frac{D'}{D}\right)^{C_P}$$

where only three out of four trimming coefficients are independent because it is also valid the below equation:

$$C_P = C_Q + C_H - C_\eta$$

5. Methodology, analysis and interpretation of results

When evaluating trimming factors, there can be two main situations:

- trimming exponents for existing or scaled pumps
- trimming exponents for completely new pumps

5.1 Trimming exponent for existing pumps

For all existing pumps there are test data at different impeller diameters, and they have been analyzed in order to extrapolate the trimming coefficients. These values are stored in both IntelliQuip ® and the merged hydraulic database and they can be used after factoring a new pump from an existing one to select the rated diameter.

5.2 Comparison of existing trimming coefficients

In general, it is possible to use the method described in [1] to estimate pump characteristics modification by impeller trimming. It considers the following equation:

$$\frac{Q'}{Q} = \left(\frac{D'}{D}\right)^m = \frac{H'}{H}$$

and the exponent m should lie roughly between 2 and 3, and it can be calculated by the below equation

$$m = 2.08 + 1.1e^{15\left(\frac{D'}{D}-1\right)}$$

Note that for impeller with under-filed blades at outlet, *m* is closer to 3 for trims smaller than the profile length. If trimming exceeds 5% of the impeller diameter, *m* is near 2. If the outlet blade angle β_2 and the outlet impeller width b_2 do not change with the radius, in theory $m = 2$ applies.

Moreover, as regards power, due to trimming the efficiency at the BEP approximately drops according to:

$$\Delta\eta = \varepsilon\left(1 - \frac{D'}{D}\right)$$

where $\varepsilon = 0.15 \div 0.25$ for volute pumps and $\varepsilon = 0.4 \div 0.5$ for diffuser pumps.

If test data are available, the value of m can be calculated as per below ratio:

$$m = \frac{ln\frac{H'}{H}}{ln\frac{D'}{D}}$$

If no test data are available, as rule of thumb for Gabbioneta heritage pumps, given that for the hydraulics of this legacy most of the times $N_q < 40$, the following coefficients have been found to work with an acceptable approximation:

$$C_Q = 1$$

$$C_H = 2$$

$$C_\eta = 0.15 \div 0.3$$

$$C_P = 2.85 \div 2.7$$

considering lower limit coefficient of efficiency for volute pumps and upper limit values for diffuser pumps.

As regards TMP legacy the following empirical correlations can be considered:

$$C_Q = 1.384 + 0.00489\, N_q$$

$$C_H = 2.2096 - 0.00117\, N_q$$

with $N_q = rpm\, \frac{\sqrt{Q/f_q}}{H^{0.75}}$

where:

Q is BEP flowrate (m^3/s)

H is pump head at BEP

f_q 1 or 2 respectively for single and double suction impellers

Below Figure 1 and Figure 2 show a comparison of the three formulations indicated above for the trimming coefficients. As regards the method referenced in [1] by Gulich, *m* value ($=C_Q=C_H$) for D'/D = 0.95 is shown, otherwise a band between 2 and 3 should be highlighted.

Both TPI and TMP methods use less conservative coefficients, meaning that less drop in flow and head is foreseen due to trimming. So, mainly when selecting the proper impeller rated diameter with these coefficients at internal performance tests rather than when issuing proposal PPCs, it is important to start from a diameter close to the trimmed final one and move forward carefully with succeeding trims, especially for high Nq (> 40 SI units) if no previous test data are available. This is to avoid the risk of trimming too much an impeller going out of foreseen API 610 test tolerances for rated point (see Table 16 [3]).

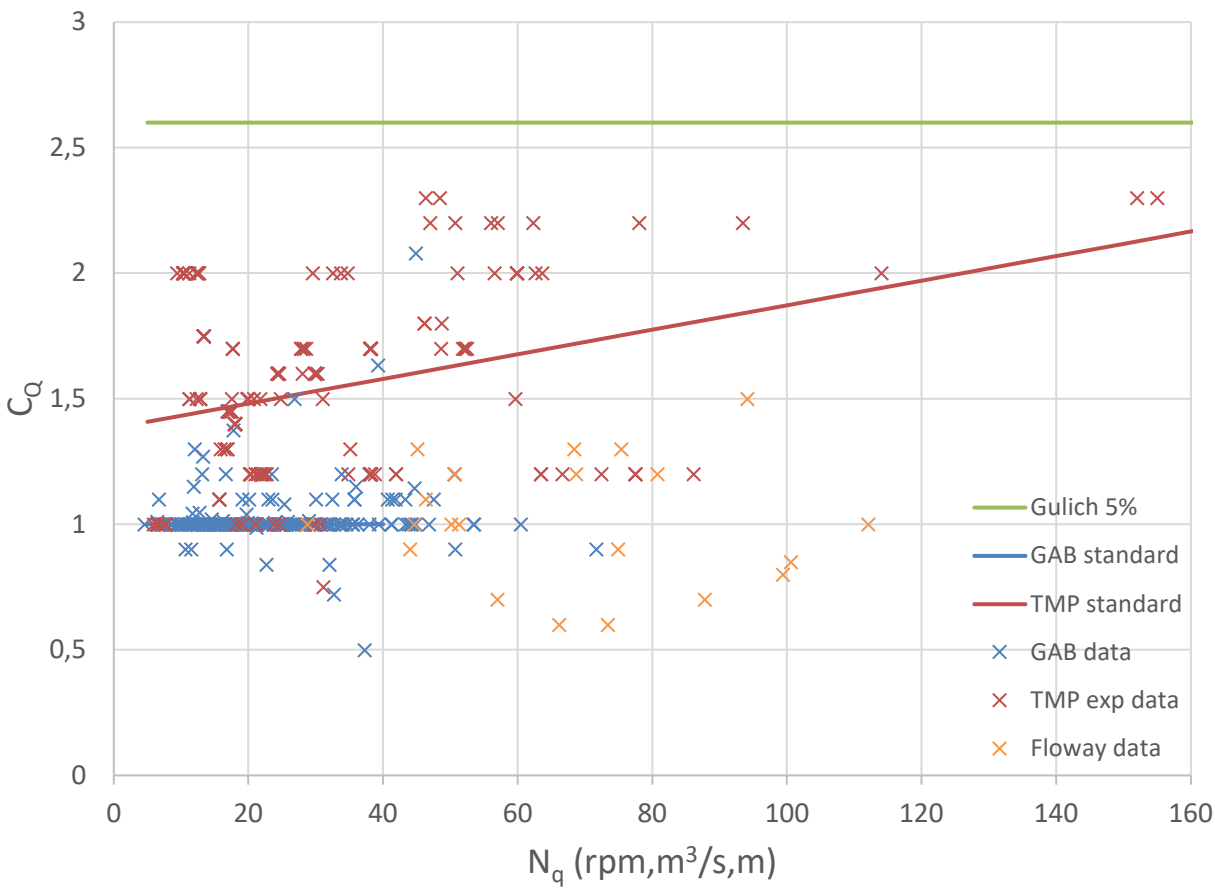


Figure 1. Flow modification coefficient for impeller trimming, comparison between methods

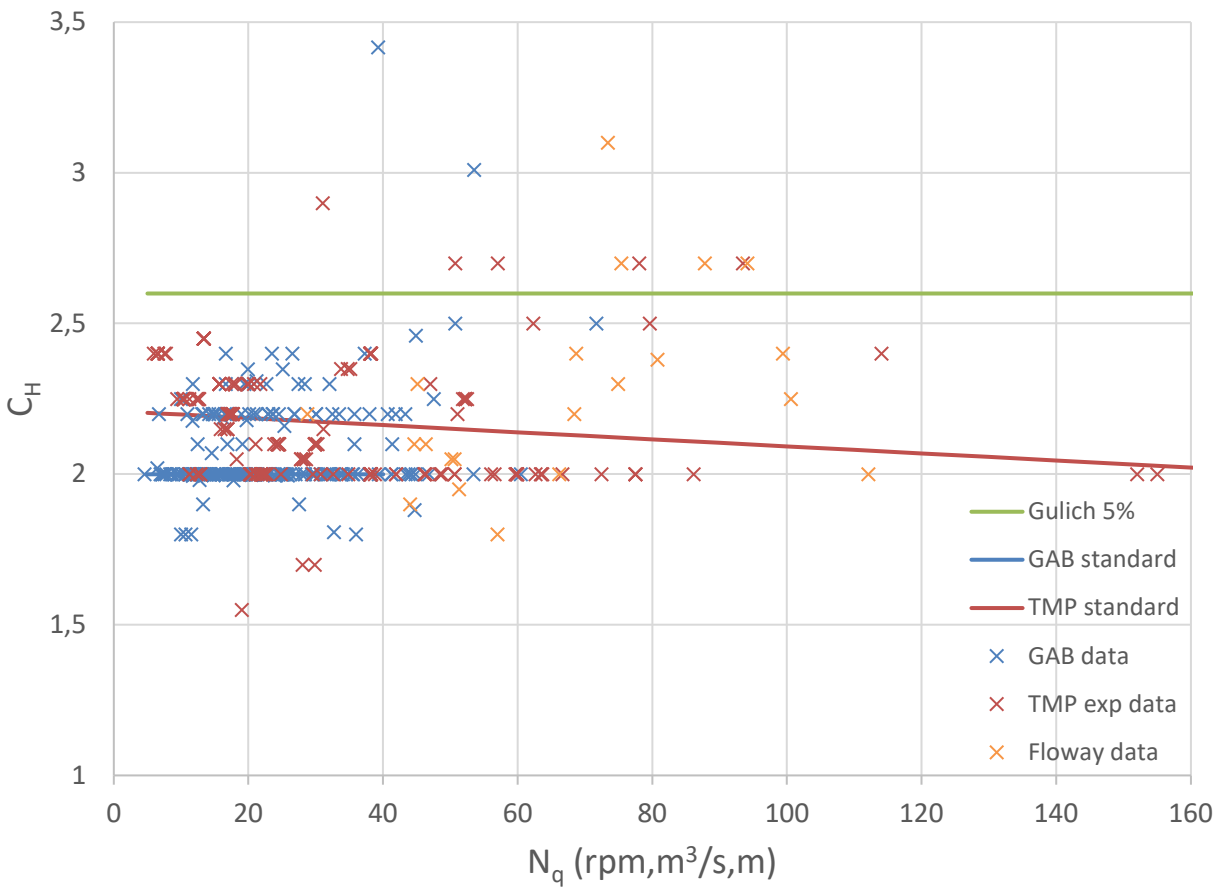


Figure 2. Head modification coefficient for impeller trimming, comparison between methods

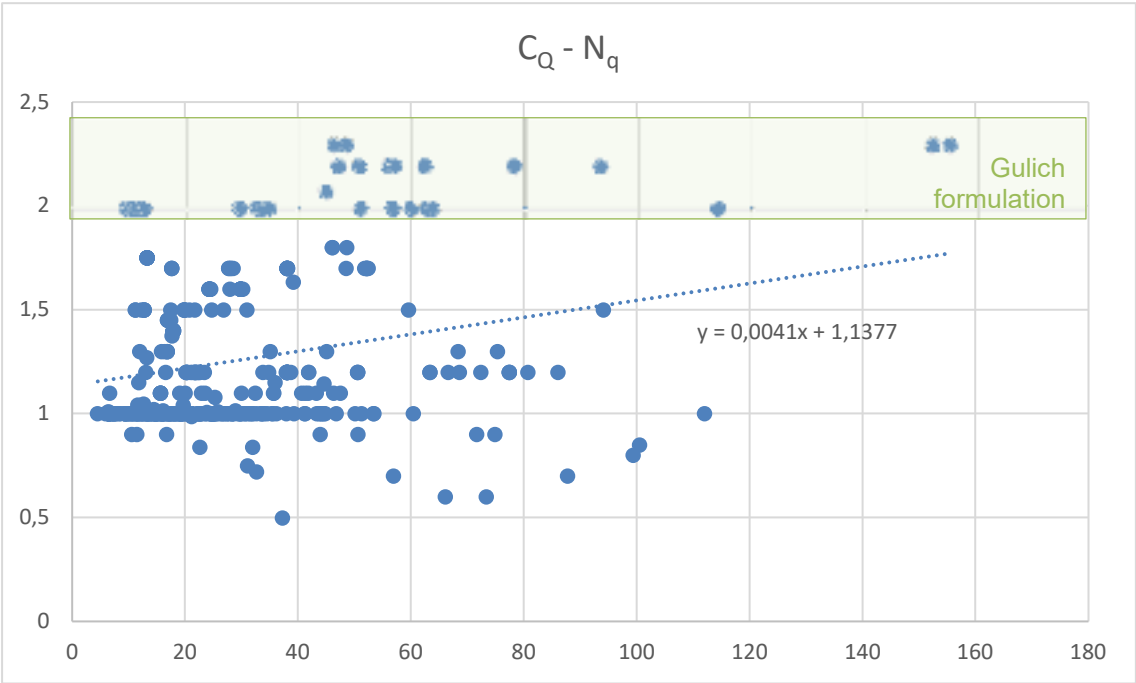
The graphs include also all the data obtained from the hydraulics of the three legacies: Gabbioneta, Termomeccanica and Floway.

5.3 Trimming exponent for new pumps

For new pumps a linear regression has been calculated considering all the data available and it can be used instead of the more general method by Gulich:

$$C_Q = 1.1377 + 0.004077\ N_q$$

$$C_H = 2.0953 + 0.001307\ N_q$$



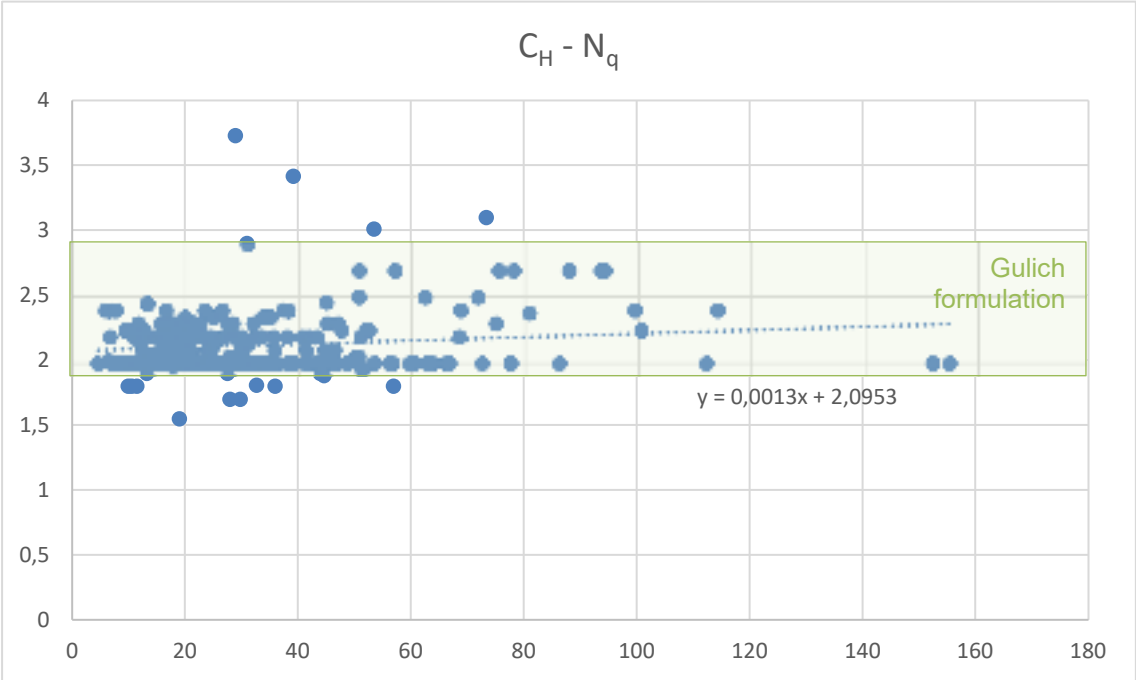


Figure 3. Linear regression considering all data from hydraulics of Gabbioneta, Termomeccanica and Floway legacies

From Figure 3 it is possible to understand that also this extrapolated final correlation is quite different from the Gulich formulation, especially for Cq, and it predicts lower trimming diameters.

6. Acceptance Criteria

It is always preferred to use pump characteristics modification coefficients obtained from testing data. But if not available, for new pumps, it is possible to use the coefficients extracted from all hydraulics data available through the linear regression of Figure 3. Otherwise, it is possible to use the more general empirical method here indicated from reference [1] which gives more conservative coefficients.

See also SOP-475 [2] for a complete explanation of method for refiguring PPC from tested data.

7. References

[1] Centrifugal Pumps, Fourth Edition by Johann Friedrich Gulich, Springer Nature Switzerland AG 2020

[2] SOP-475 – Method for Refiguring Pump Performance from Tested Data

[3] Table 16 – Performance tolerances, *API STANDARD 610, 12th Edition, January 2021*